NatBool, $\Rightarrow$, Nat, $0, 1, +, \times$, generikus programozás = Fin+mai

$>2 : \mathrm{Nat} \to \mathrm{Bool}$ generikus map

$\mathrm{Nat} \times \mathrm{Bool} \longrightarrow \mathrm{Bool} \times \mathrm{Bool}$

$\mathrm{Nat} + \mathrm{Bool} \longrightarrow \mathrm{Bool} + \mathrm{Bool}$

$\boxed{\mathrm{Nat} + \mathrm{Nat}} \longrightarrow \mathrm{Bool} + \mathrm{Bool}$

$\underbrace{\mathrm{Nat} \times \ldots \times \mathrm{Nat}}_{\omega \text{ sokszor}, \ \infty \text{ sokszor}} \overset{?}{=} \mathrm{Nat}^{\omega} \overset{?}{=} \mathrm{Nat} \Rightarrow \mathrm{Nat} \longrightarrow \mathrm{Nat} \Rightarrow \mathrm{Bool}$

$A \Rightarrow \mathrm{Nat} \longrightarrow A \Rightarrow \mathrm{Bool}$

$\boxed{\mathrm{Nat} + \mathrm{Nat}} \longrightarrow \mathrm{Nat} + \mathrm{Bool}$

$\mathrm{map} : (a \to b) \to [a] \to [b]$

$\mathrm{map} : (a \to b) \to F a \to F b$

típusoperátor

$F : \mathrm{Ty} \to \mathrm{Ty}$

$\mathrm{map} : (F : \mathrm{Ty} \to \mathrm{Ty}) \to (\mathrm{Tm}\,A \to \mathrm{Tm}\,B) \to \mathrm{Tm}(FA) \to \mathrm{Tm}(FB)$

$A = \mathrm{Nat}, B = \mathrm{Bool}$

$FX = X \times \mathrm{Bool}$ $\quad FA = \mathrm{Nat} \times \mathrm{Bool}$ $\quad FB = \mathrm{Bool} \times \mathrm{Bool}$

$FX = X + \mathrm{Bool}$

$FX = X + X$

$FX = A \Rightarrow X$

$FX = \mathrm{Nat} + X$

$FX = \mathrm{Nat} \Rightarrow X$

$\mathrm{map}\,(\lambda X. X \times \mathrm{Bool})\ \overset{: \mathrm{Tm}\,A \to \mathrm{Tm}\,B}{f}\ \underset{: \mathrm{Tm}(A \times \mathrm{Bool})}{w}$

$: \mathrm{Tm}(B \times \mathrm{Bool})$

$:= (f(\mathrm{fst}\,w), \mathrm{snd}\,w)$

$\mathrm{map}\,(\lambda X. C \Rightarrow X)\ \overset{: \mathrm{Tm}\,A \to \mathrm{Tm}\,B}{f}\ \overset{: \mathrm{Tm}(C \Rightarrow A)}{w} : \mathrm{Tm}(C \Rightarrow B)$

$:= \mathrm{lam}\,\lambda c.\ f(\underbrace{w \cdot c}_{: \mathrm{Tm}\,A})$

ez OK

$FX = \top \Rightarrow \bot$

$\mathrm{map}\,F\ \underset{: \mathrm{Tm}\,A \to \mathrm{Tm}\,B}{f}\ w = w$

$: \mathrm{Tm}(FA) = \mathrm{Tm}(\top \Rightarrow \bot)$

nem jó

$FX = X \Rightarrow \bot$

$\mathrm{map}\,F\,f\,(w : \mathrm{Tm}(A \Rightarrow \bot)) : \mathrm{Tm}(B \Rightarrow \bot)$

$$\begin{tikzcd}
A \arrow[r, "f"] \arrow[d, "w"'] & B \arrow[dl, dotted, "?"] \\
\bot &
\end{tikzcd}$$

nem jó

$FX = X \Rightarrow X$

$$\begin{tikzcd}
A \arrow[r, "f"] \arrow[d, "w"'] & B \arrow[d, "?"] \\
A \arrow[r, "f"] & B
\end{tikzcd}$$

$\mathrm{map}\,F\,(\lambda x. x)\,w = w$

$\mathrm{map}\,F\,(\lambda x. f(g\,x))\,w = \mathrm{map}\,F\,f\,(\mathrm{map}\,F\,g\,w)$

---

1. polinom, 2. pozitív, 3. neg. poz.

F polinom, ha pl. $FX = 1 + 2X^2 + 3X^4$

$FX = \top + \mathrm{Bool} \times X \times X + (\mathrm{Bool} + \top) \times X \times X \times X \times X$

$\mathrm{Ty}, \mathrm{Tm},$

$\mathrm{Poly} : (\mathrm{Ty} \to \mathrm{Ty}) \to \mathrm{Set}$

$\_ : \mathrm{Poly}(\lambda X. \top)$

$\_ : \mathrm{Poly}(\lambda X. \bot)$

$\_ : \mathrm{Poly}(\lambda X. X)$

$\_ : \mathrm{Poly}\,F \to \mathrm{Poly}\,G \to \mathrm{Poly}(\lambda X. FX \times GX)$

$\_ : \mathrm{Poly}\,F \to \mathrm{Poly}\,G \to \mathrm{Poly}(\lambda X. FX + GX)$

pl. $p : \mathrm{Poly}(\lambda X. X \times \mathrm{Bool})$

$$\dfrac{\dfrac{}{\mathrm{Poly}(\lambda X. X)} \qquad \dfrac{\dfrac{}{\mathrm{Poly}(\lambda X. X)} \quad \dfrac{}{\mathrm{Poly}(\lambda X. \top)}}{\mathrm{Poly}(\lambda X. X \times \top)}}{\mathrm{Poly}(\lambda X. \underset{FX}{X} + \underset{GX}{X \times \top})}$$

$\mathrm{map} : (F : \mathrm{Ty} \to \mathrm{Ty}) \to \{\mathrm{Poly}\,F\} \to (\mathrm{Tm}\,A \to \mathrm{Tm}\,B) \to \mathrm{Tm}(FA) \to \mathrm{Tm}(FB)$

$\mathrm{map}\,(\lambda X. \top)\,f\,w = \cancel{w}\ \ \mathrm{tt}$ $\quad \leftarrow w : \underbrace{\mathrm{Tm}(FA)}_{} = \mathrm{Tm}((\lambda X. \top) A) = \mathrm{Tm}\,\top$

$\mathrm{map}\,(\lambda X. \bot)\,f\,w = \cancel{w}\ \ \mathrm{exfalso}\,w$

$\mathrm{map}\,(\lambda X. X)\,f\,w = f\,w$

$\mathrm{map}\,(\lambda X. FX \times GX)\,f\,\underset{: \mathrm{Tm}(FA \times GA)}{w} = (\mathrm{map}\,F\,f\,(\mathrm{fst}\,w), \mathrm{map}\,G\,f\,(\mathrm{snd}\,w))$ $\quad : \mathrm{Tm}(FB \times GB)$

$\mathrm{map}\,(\lambda X. FX + GX)\,f\,\underset{: FA + GA}{w} = \mathrm{case}\,w$ $\quad FB + GB$

$\quad (\lambda \underset{: FA}{w_A}. \mathrm{inl}(\underbrace{\mathrm{map}\,F\,f\,w_A}_{FB}))$ $\quad : FB + GB$

$\quad (\lambda \underset{: GA}{w_B}. \mathrm{inr}(\underbrace{\mathrm{map}\,G\,f\,w_B}_{GB}))$ $\quad FB + GB$

$\mathrm{map}\,F\,\underbrace{\{p_F\}}_{: \mathrm{Poly}\,F}\,\mathrm{id}\,w = w$

$\mathrm{map}\,(\lambda X. \top)\,\mathrm{id}\,w = \mathrm{tt} \overset{\top\eta}{=} w$

$\mathrm{map}\,(\lambda X. \bot)\,\mathrm{id}\,\underset{: \mathrm{Tm}\,\bot}{w} = \underbrace{\mathrm{exfalso}\,w}_{: \mathrm{Tm}\,\bot} \overset{\bot\eta}{=} \mathrm{id}\,w = w$

$\top\eta : (t : \mathrm{Tm}\,\top) \to t = \mathrm{tt}$

$\bot\eta : (f : \mathrm{Tm}\,\bot \to \mathrm{Tm}\,A) \to f = \mathrm{exfalso}$

$\mathrm{map}\,(\lambda X. X)\,\mathrm{id}\,w \overset{\mathrm{def}}{=} \mathrm{id}\,w \overset{\mathrm{def}}{=} w$

$\mathrm{map}\,(\lambda X. FX \times GX)\,\mathrm{id}\,w = (\mathrm{map}\,F\,\mathrm{id}\,(\mathrm{fst}\,w), \mathrm{map}\,G\,\mathrm{id}\,(\mathrm{snd}\,w))$

i.h.$_1$: $\mathrm{map}\,F\,\mathrm{id}\,w' = w'$

i.h.$_2$: $\mathrm{map}\,G\,\mathrm{id}\,w'' = w''$

$\quad \| $

$(\mathrm{fst}\,w, \mathrm{snd}\,w)$

$\quad \| \times\eta$

w

$\mathrm{map}\,(\lambda X. FX + GX)\,\mathrm{id}\,w = \mathrm{case}\,w$

i.h.$_{1,2}$ ugyanaz

$\quad (\lambda w_F. \mathrm{inl}(\mathrm{map}\,F\,\mathrm{id}\,w_F))$

$\quad (\lambda w_G. \mathrm{inr}(\mathrm{map}\,G\,\mathrm{id}\,w_G))$

$\quad \| $ i.h.

$\mathrm{case}\,w\,(\lambda w_F. \mathrm{inl}\,w_F)\,(\lambda w_G. \mathrm{inr}\,w_G)$

$\quad \|$

$\mathrm{case}\,w\,\mathrm{inl}\,\mathrm{inr}$

$\quad \| +\eta$

w

$+\eta : (f : \mathrm{Tm}(A+B) \to \mathrm{Tm}\,C)$

$\to f = \mathrm{case}\,\_\,(f \circ \mathrm{inl})\,(f \circ \mathrm{inr})$

$\boxed{f = \mathrm{id}} \mapsto \mathrm{id} = \mathrm{case}\,\_\,\mathrm{inl}\,\mathrm{inr}$

1. map polinomiális típusoperátorokra

---

3 map neg. poz. típusop-ra $=$ ① $+$

$\_ : \mathrm{Poly}\,F \to \mathrm{Poly}(\lambda X. A \Rightarrow FX)$

$\lambda X. 1 + X^2 + 3X^1$

$\mathrm{Poly}(\lambda X. \top + (\mathrm{Bool} \Rightarrow X) + (\top + \top + \top) \times (\top \Rightarrow X))$

$\mathrm{map}\,(\lambda X. \underline{C \Rightarrow FX})\,f\,\underline{w} := \mathrm{lam}\,\lambda c.\,\mathrm{map}\,F\,f\,(\underbrace{w \cdot c}_{: FA})$

$: \mathrm{Tm}(C \Rightarrow FA)$ $\quad : \mathrm{Tm}(C \Rightarrow FB)$

$\underbrace{FX \times FX \times \ldots \times FX}_{C\text{-szer}}$

$\mathrm{Poly}(\lambda X. \mathrm{Bool} \Rightarrow (\mathrm{Bool} \Rightarrow X))$

$\lambda X. (X^2)^2$

id szabály: $\mathrm{lam}\,\lambda c. w \cdot c \overset{\Rightarrow\eta}{=} w$

$\lambda X. A \Rightarrow X$

$$\dfrac{\dfrac{\lambda X. X}{\mathrm{Poly}} \qquad \dfrac{\lambda X. \top}{\mathrm{Poly}}}{\mathrm{Poly} \quad \times} \qquad \lambda X. \bot$$

$FX = (X \Rightarrow C) \Rightarrow X$ $\quad \rightarrow$ `data F x = Mk ((x -> c) -> x)` `deriving (Functor)`

input $\qquad$ output

$$\begin{tikzcd}
A \arrow[r, "f"] & B \arrow[d, "h"] \\
& C
\end{tikzcd}$$

$g : (A \to C) \to A$ $\qquad \boxed{? : B}$

$g(h \circ f) : A$

$f(g(h \circ f)) : B$

$$\dfrac{\dfrac{\dfrac{}{\mathrm{Pos}(\lambda X. X)} \qquad \dfrac{}{\mathrm{Neg}(\lambda X. C)}}{\mathrm{Neg}(\lambda X. X \Rightarrow C)} \qquad \dfrac{}{\mathrm{Pos}(\lambda X. X)}}{\mathrm{Pos}(\lambda X. (X \Rightarrow C) \Rightarrow X)}$$

$\mathrm{Pos}, \mathrm{Neg} : (\mathrm{Ty} \to \mathrm{Ty}) \to \mathrm{Set}$

$\mathrm{Pos}(\lambda X. C)$ $\qquad \mathrm{Neg}(\lambda X. C)$

$\mathrm{Pos}(\lambda X. X)$

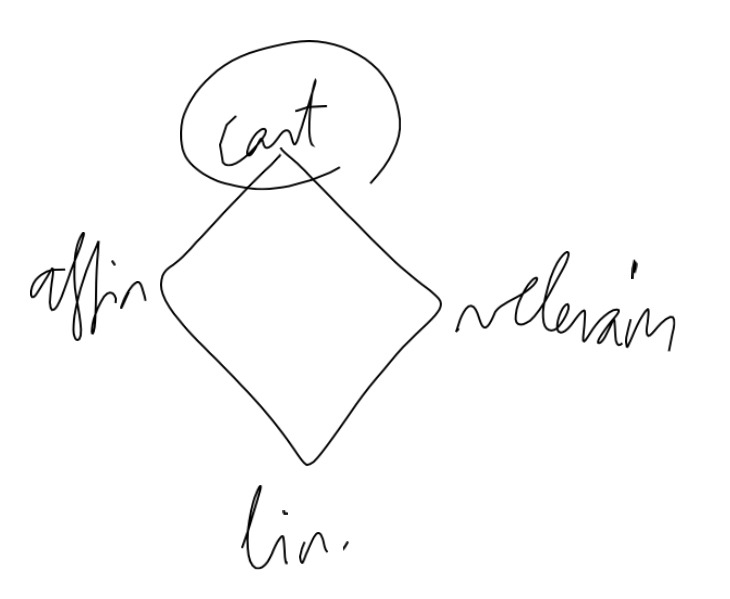
$\mathrm{Pos}\,F \to \mathrm{Pos}\,G \to \mathrm{Pos}(\lambda X. \begin{cases} FX + GX \\ FX \times GX \end{cases})$ $\qquad \mathrm{Neg}\,F \to \mathrm{Neg}\,G \to \begin{cases} \mathrm{Neg}(\lambda X. FX + GX) \\ \mathrm{Neg}(\lambda X. FX \times GX) \end{cases}$

$\mathrm{Neg}\,F \to \mathrm{Pos}\,G \to \mathrm{Pos}(\lambda X. FX \Rightarrow GX)$

$\mathrm{Pos}\,F \to \mathrm{Neg}\,G \to \mathrm{Neg}(\lambda X. FX \Rightarrow GX)$

$$\dfrac{\dfrac{}{\mathrm{Pos}(\lambda X. X)} \qquad \dfrac{}{\mathrm{Neg}(\lambda X. \mathrm{Bool})}}{\mathrm{Neg}(\lambda X. \underset{P}{X} \Rightarrow \underset{N}{\mathrm{Bool}})} \qquad \dfrac{\dfrac{?}{\mathrm{Neg}(\lambda X. X)} \qquad \dfrac{✓}{\mathrm{Pos}(\lambda X. \mathrm{Bool})}}{\mathrm{Pos}(\lambda X. X \Rightarrow \mathrm{Bool})}$$

$$\begin{tikzcd}
B \arrow[r, "f"] \arrow[dr, "?"'] & A \arrow[d, "w"] \\
& \mathrm{Bool}
\end{tikzcd}$$



$\mathrm{map}^P : (F : \mathrm{Ty} \to \mathrm{Ty}) \to \{\mathrm{Pos}\,F\} \to (\mathrm{Tm}\,A \to \mathrm{Tm}\,B) \to \mathrm{Tm}(FA) \to \mathrm{Tm}(FB)$

$\mathrm{map}^N : (F : \mathrm{Ty} \to \mathrm{Ty}) \to \{\mathrm{Neg}\,F\} \to (\mathrm{Tm}\,B \to \mathrm{Tm}\,A) \to \mathrm{Tm}(FA) \to \mathrm{Tm}(FB)$

$\mathrm{map}^P\,(\lambda X. C)\,f\,w = w$

$\mathrm{map}^P\,(\lambda X. X)\,f\,w = f\,w$

$\mathrm{map}^P\,(\lambda X. FX \times GX)\,f\,w = (\mathrm{map}^P\,F\,f\,(\mathrm{fst}\,w), \mathrm{map}^P\,G\,f\,(\mathrm{snd}\,w))$

$\mathrm{map}^N\,(\lambda X. C)\,f\,w = w$

$\mathrm{map}^N\,(\lambda X. FX \times GX)\,f\,w = (\mathrm{map}^N\,F\,f\,(\mathrm{fst}\,w), \mathrm{map}^N\,G\,f\,(\mathrm{snd}\,w))$

$\mathrm{map}^P\,(\lambda X. FX \Rightarrow GX)\,f\,w = \mathrm{lam}\,\lambda x.\,\mathrm{map}^P\,G\,f\,(w \cdot \mathrm{map}^N\,F\,f\,x)$

$: \mathrm{Tm}(FB \Rightarrow GB)$

$A \xrightarrow{f} B$

$$\begin{tikzcd}
FA \arrow[r, "w"] & GA \arrow[d, "\mathrm{map}^P\,G\,f"] \\
FB \arrow[u, "\mathrm{map}^N\,F\,f"] \arrow[r, dotted, "?"'] & GB
\end{tikzcd}$$

$\mathrm{map}^N\,(\lambda X. FX \Rightarrow GX)\,f\,w = \mathrm{lam}\,\lambda \underset{: \mathrm{Tm}(FB)}{x}.\,\mathrm{map}^N\,G\,f\,(w \cdot \mathrm{map}^P\,F\,f\,x)$

$B \xrightarrow{f} A$

$$\begin{tikzcd}
FA \arrow[r, "w"] & GA \arrow[d, "\mathrm{map}^N\,G\,f"] \\
FB \arrow[u, "\mathrm{map}^P\,F\,f"] \arrow[r, dotted, "?"'] & GB
\end{tikzcd}$$

nem pos, neg: $FX = X \Rightarrow X$

$$\dfrac{\mathrm{Neg}(\lambda X. X) \qquad \mathrm{Pos}(\lambda X. X)}{\mathrm{Pos}(\lambda X. X \Rightarrow X)}$$

$$\begin{tikzcd}
A \arrow[r, "f"] & B \arrow[d, dotted, "?"] \\
A \arrow[u, "u"] & B \arrow[l, "f^{-1}"]
\end{tikzcd}$$